

Ltp ~ 10

Biomolecules

- ⇒ Biomolecules may be defined as the complex lifeless organic substances which form the basis of life and are responsible for the growth and maintenance of the living systems.
- ⇒ They play an active role in the process of metabolism.
- ⇒ Examples are: Carbohydrates, Proteins, fats and nucleic acids.

CARBOHYDRATES

- ⇒ Carbohydrates are optically active polyhydroxy aldehydes or polyhydroxy ketones or large polymeric molecules, which on hydrolysis yield polyhydroxy aldehydes or polyhydroxy ketones.
- ⇒ Examples are: Glucose, fructose, sucrose, starch
- ⇒ General Formula is $C_x (H_2O)_y$

Classification of Carbohydrates.

1. Monosaccharides: Monosaccharides are those polyhydroxy aldehydes or polyhydroxy ketones which cannot be decomposed further by hydrolysis to give simpler carbohydrates.

=> General Formula is $(CH_2O)_n$.

=> Examples are: Glucose ($C_6H_{12}O_6$),
Fructose ($C_6H_{12}O_6$), galactose ($C_6H_{12}O_6$),
ribose ($C_5H_{10}O_5$)

2. Oligosaccharides: Carbohydrates which upon hydrolysis give a definite number (2-10) of molecules of monosaccharides.

=> Types are:

— Disaccharides: Sucrose, Maltose, Lactose.

— Trisaccharides: Raffinose

3. Polysaccharides: Carbohydrates which upon hydrolysis give large number of monosaccharide molecules.

=> Examples are: Starch, Cellulose, Glycogen

* Classification on Taste

1. Sugars: Carbohydrates which are sweet in taste are called Sugars.

eg:- Glucose, Fructose, sucrose, lactose.

2. Non-Sugars: Carbohydrates which are not sweet in taste are called non-sugars.

eg:- Starch, Cellulose.

* Classification on Reducing Behaviour.

1. Reducing Sugars: Carbohydrates which are capable of reducing Tollens reagent and Fehlings Solution are called reducing sugars.

eg:- All mono and di-saccharides except sucrose.

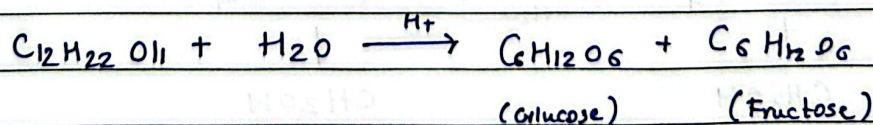
2. Non-Reducing Sugars: Carbohydrates which are unable to reduce Tollens reagent and Fehlings Solution are non-reducing sugars.

eg:- Sucrose.

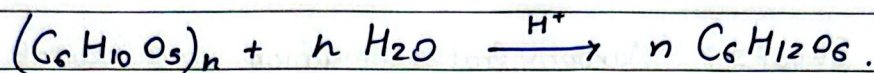
GLUCOSE

⇒ Preparation of Glucose

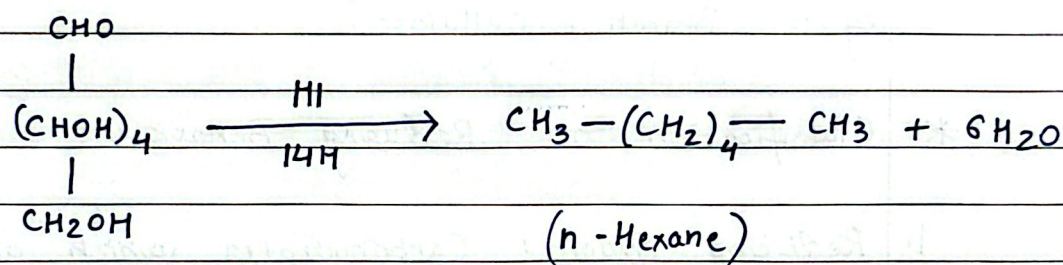
1. From Sucrose



2. From Starch

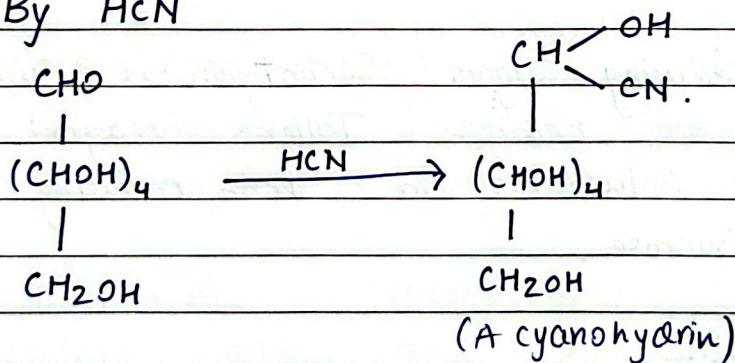
 $\Rightarrow$ Structure of Glucose

i) Presence of a straight chain of six carbon atoms:

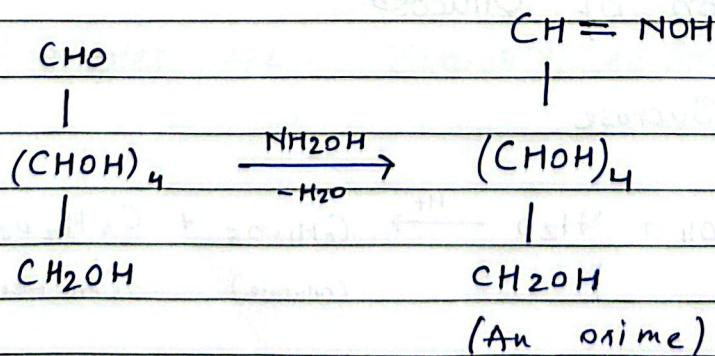


ii) Presence of a carbonyl group.

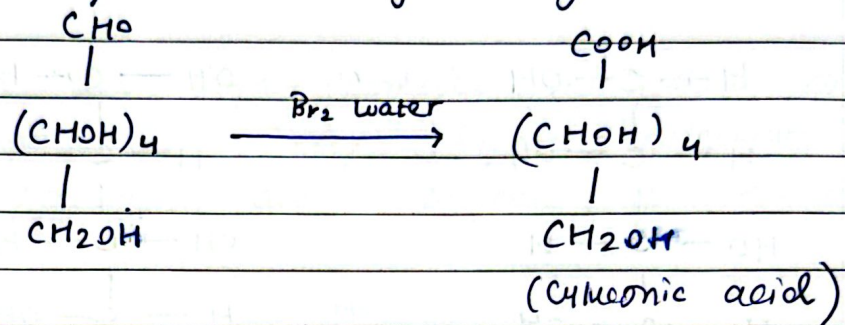
— By HCN



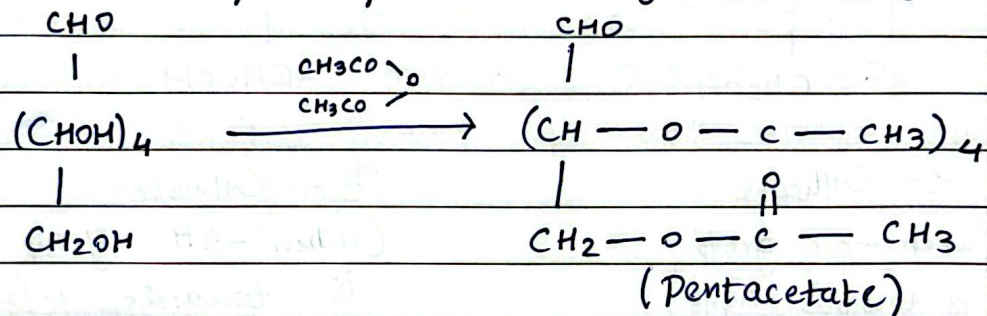
— By NH_2OH



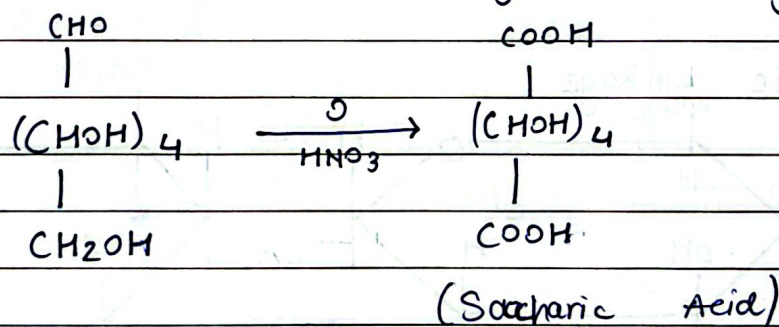
iii] Presence of an Aldehydic group.



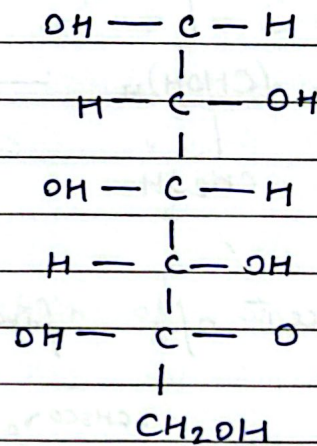
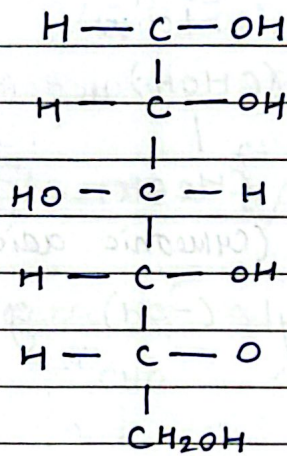
iv] Presence of five hydroxyl (-OH) groups



v] Presence of a primary alcohol group.



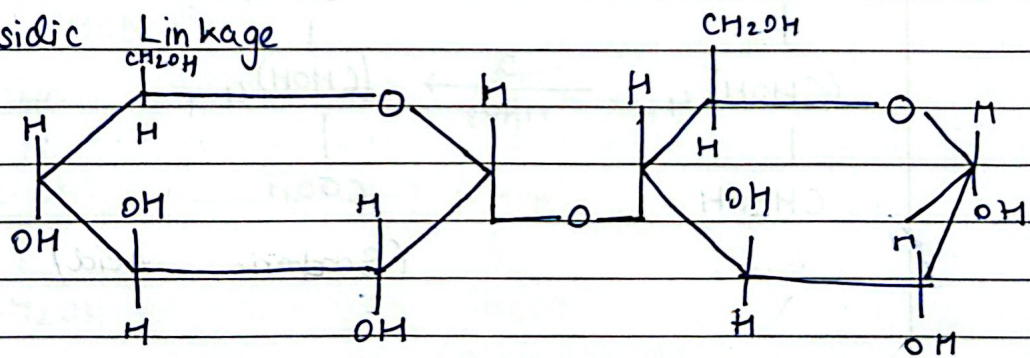
=> Structure of Glucose.



α -Glucose
(when -OH group
is towards right)

β -Glucose
(when -OH group
is towards left)

=> Glycosidic Linkage

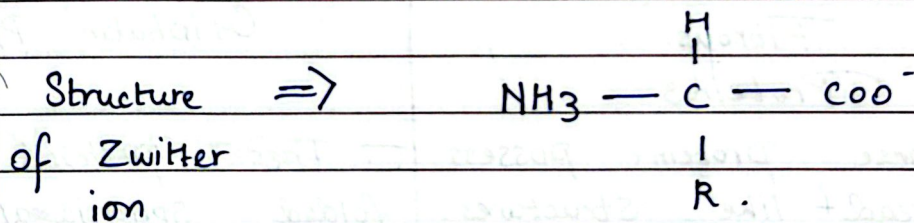
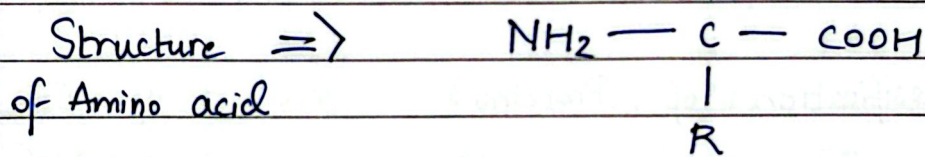


PROTEINS

⇒ Proteins are highly complex nitrogenous organic compound of very high molecular masses. All proteins are polymers of α -amino acids.

AMINO ACIDS

⇒ Amino acids are the basic building units of Proteins. Their molecules possess both an amino ($-\text{NH}_2$) group as well as a carboxylic ($-\text{COOH}$) group.



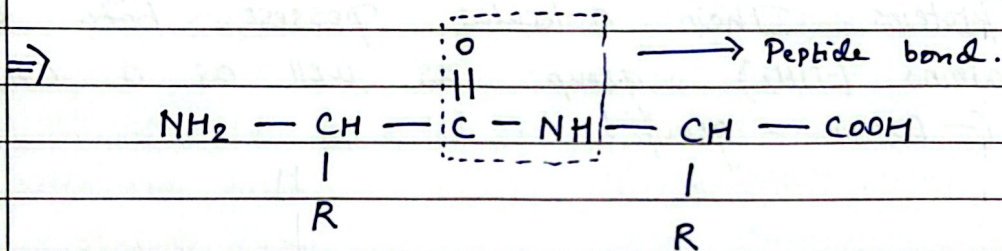
⇒ Classification of Amino Acids.

- i) Neutral - when $-\text{NH}_2$ group atoms and $-\text{COOH}$ group atoms are equal
- ii) Acidic - when $-\text{COOH}$ group atoms are more than $-\text{NH}_2$ groups
- iii) Basic - when $-\text{NH}_2$ group atoms are more than $-\text{COOH}$ group.

PEPTIDES.

⇒ The compounds obtained by condensation of two or more, α -amino acids are called Peptides.

⇒ Peptide bond is the bond formed by the condensation of two amino acids molecule.



⇒ Classification of Proteins.

Fibrous Proteins	Globular Proteins
- These proteins possess thread-like structures. - These proteins are insoluble in water. - These are stable to moderate changes in temperature and pH.	- These proteins possess folded spheroidal structures. - These proteins are soluble in water. - These are very sensitive to changes in temperature and pH.

⇒ Denaturation of Proteins.

The process which leads to a change in physical and biological properties of a protein without affecting its chemical composition is called denaturation.

⇒ Methods.

- i] By heat. - Egg.
- ii] By Alcohol -
- iii] By conc. acids -
- iv] By salts of heavy metals.
- v] By alkaloidal reagents.
- vi] By radiation.

⇒ Examples are: boiling of egg and preparation of cheese from milk.

* NUCLEIC ACIDS

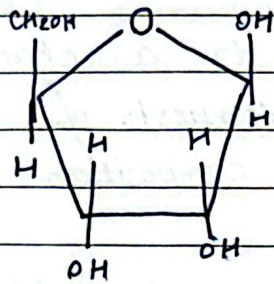
⇒ Nucleic acids constitute an important class of biomolecules which are present in the nuclei of all living cells.

⇒ Nucleic acids are made up of nucleotides, they are actually biopolymers.

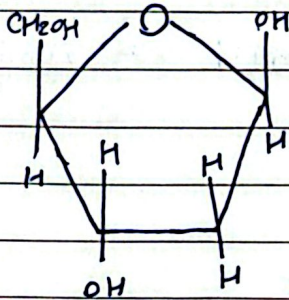
⇒ Types of Nucleic acids.

- i] Deoxyribonucleic acid (DNA)
- ii] Ribonucleic acid (RNA)

⇒ Structure of Nucleic acids.



⇒ Ribonucleic acid.
(RNA)



⇒ Deoxyribonucleic acid
(DNA).

⇒ Purines :

Adenine (A) & Guanine (G)

⇒ Pyrimidines :

— Thymine (T), Cytosine (C) & Uracil (U)

RNA :	Uracil
	Adenine
	Guanine
	Cytosine
DNA :	Cytosine
	Adenine
	Guanine
	Thymine

=> NUCLEOSIDES :

BASE + SUGAR = Nucleoside.

=> NUCLEOTIDE :

BASE + SUGAR + PHOSPHATE = Nucleotide.

* MUTATIONS

=> A chemical change in a DNA molecule which could lead to synthesis of proteins with a different amino acid sequence is termed as mutation.

* VITAMINS

=> Vitamins may be defined as a group of organic compounds which are required in very small amounts for the healthy growth and normal functioning of animal organisms.

=> i] Fat - Soluble Vitamins : Vitamins A, D, E and K.

ii] Water - Soluble Vitamins : Vitamin B and C.

Vitamin H → exception

Vitamins, Sources and Deficiency Table

	Vitamins	Sources	Deficiency Diseases
1.	Vitamin A	Milk, eggs, fish tomatoes	Night blindness, Xerosis, Xerophthalmia.
2.	Vitamin B ₁	milk, eggs, green vegetables	Beri Beri and loss of appetite
3.	Vitamin B ₂	Milk, cheese, tomatoes	Cheilosis, Retardation of growth and inflammation of tongue.
4.	Vitamin B ₃	Milk, eggs, -tomatoes	Dermatitis, greying of hair, retardation of body and mental health
5.	Vitamin B ₆	Milk, eggs, yeast fish etc.	Anaemia, Insomnia Pellagra, general weakness.
6.	Vitamin B ₁₂	Milks, eggs, meat liver etc	Pernicious anaemia and inflammation of tongue mouth.
7.	Vitamin C	Citrus fruits tomatoes, chillies	Scurvy, skin diseases.
8.	Vitamin D	milk, eggs, liver	Rickets, Deformation of bones and teeth
9.	Vitamin E	Milk, oils, wheat fish.	Sterility, muscular atrophy
10.	Vitamin K	Cabbage, eggs, liver colon	Haemorrhage, lengthening of the time of blood clotting